Fundamental Physics 1: Mechanics

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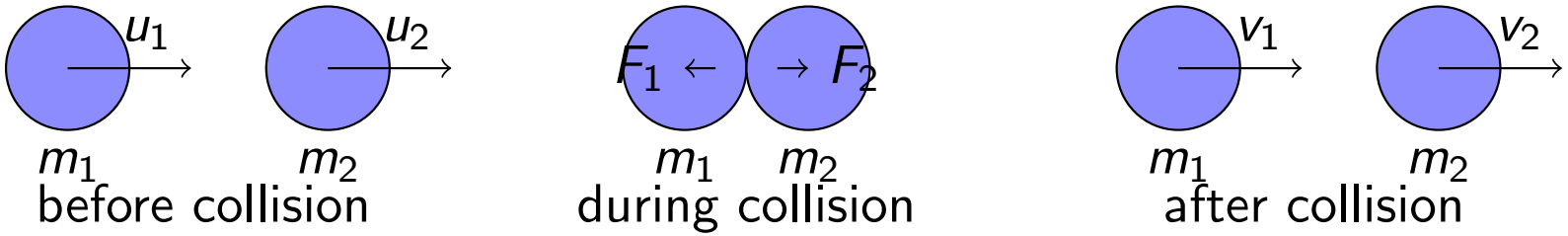
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A collision is a phenomenon which has these features:

1. it occurs in a short time interval,
2. what happens after the collision differs from what happens before the collision,
3. the colliding bodies may be assumed to constitute a closed system,
4. momentum and energy are conserved during the collision.

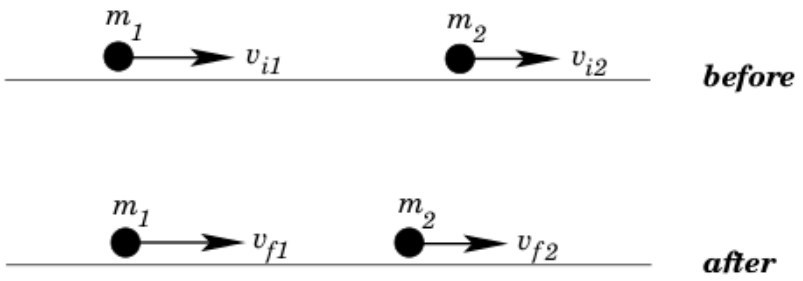


Vector sum of linear momentum after collision = vector sum of linear momentum before collision

$\Rightarrow$ **Linear momentum is conserved!**

Chapter 3: Collisions in 1-dimension

- m_1 and m_2 collide in 1-dimension
- ignore zero net external force in the direction of moving
  - Momentum conservation gives



$$m_1\vec{v}_{i1} + m_2\vec{v}_{i2} = m_1\vec{v}_{1f} + m_2\vec{v}_{f2} \quad (1)$$

  - In 1-dimention: $m_1 v_{i1} + m_2 v_{i2} = m_1 v_{f1} + m_2 v_{f2}$

Types of collision

- Elastic collision: Total kinetic energy of the two colliding objects is conserved

$$\frac{1}{2}m_1 v_{i1}^2 + \frac{1}{2}m_2 v_{i2}^2 = \frac{1}{2}m_1 v_{f1}^2 + \frac{1}{2}m_2 v_{f2}^2$$

- Inelastic collision: some fraction of the initial kinetic energy of the colliding objects is usually converted into some other form of energy.
- Totally inelastic collision: two objects stick together after the collision $\Rightarrow v_{f1} = v_{f2}$

There are many types of collision

- Elastic collision: Total kinetic energy of the two colliding objects is conserved

$$\frac{1}{2}m_1 v_{i1}^2 + \frac{1}{2}m_2 v_{i2}^2 = \frac{1}{2}m_1 v_{f1}^2 + \frac{1}{2}m_2 v_{f2}^2$$

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Newton's law of restitution

- **When two bodies collide, relative velocity after collision =** $-e\times$ **(relative velocity before collision)**.
- e ; $(0 \leq e \leq 1)$ is coefficient of restitution
- **Elastic collision:** $e = 1$
- **In-elastic collision,** $0 < e < 1$
- **Total in-elastic collision,** $e = 0$

Chapter 3: Elastic collision (1)

- Momentum conservation

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$

- Using Newton's law of restitution

$$v_1 - v_2 = -e(u_1 - u_2)$$

- Results:

$$v_1 = \frac{2m_2 u_2}{m_1 + m_2} + \frac{(m_1 - m_2)u_1}{m_1 + m_2}$$

$$v_2 = \frac{2m_1 u_1}{m_1 + m_2} + \frac{(m_2 - m_1)u_2}{m_1 + m_2}$$

$$v_1 = \frac{2m_2u_2}{m_1+m_2} + \frac{(m_1-m_2)u_1}{m_1+m_2}$$

$$v_2 = \frac{2m_1u_1}{m_1+m_2} + \frac{(m_2-m_1)u_2}{m_1+m_2}$$

We consider 3 following cases:

1. $m_1 = m_2$ In an elastic collision between two bodies of the same mass, the velocities of the bodies interchange after collision
2. $m_2 >> m_1$ and m_2 at rest
   $v_1 = -u_1$ and $v_2 = 0$
   The small mass m_1 rebounces the same speed, whereas the large mass m_2 remains at rest.
3. m_1 is very large compared to m_2 ($m_1 >> m_2$) and m_2 initially at rest.

$$v_1 = u_1, \quad \text{and} \quad v_2 = 2u_1$$

- We have: $v_f = v_{f1} = v_{f2}$
- $\Rightarrow m_1 v_{i1} + m_2 v_{i2} = m_1 v_{f1} + m_2 v_{f2} = (m_1 + m_2) v_f$

$$\Rightarrow v_f = \frac{m_1 v_{i1} + m_2 v_{i2}}{m_1 + m_2} = v_{cm}$$

two objects remain stationary in the center of mass velocity frame.

- Suppose that $v_{i2} = 0$

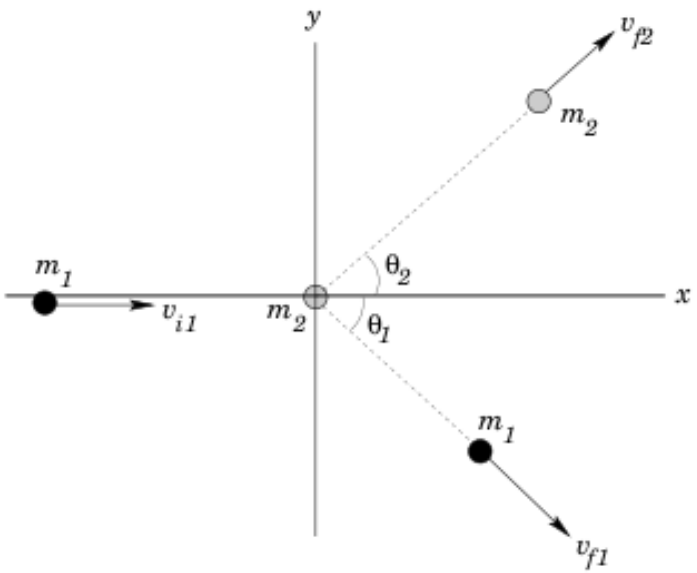
$$v_f = \frac{m_1}{m_1 + m_2} v_{i1}$$

$\rightarrow$ The fractional loss in kinetic energy of the system due to the collision is

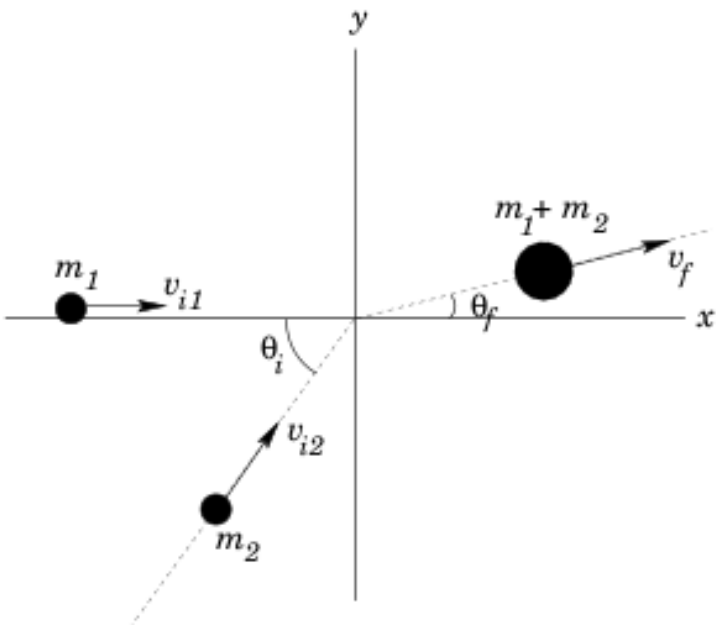
$$f = \frac{K_i - K_f}{K_i} = \frac{m_1 v_{i1}^2 - (m_1 + m_2) v_f^2}{m_1 v_{i1}^2} = \frac{m_2}{m_1 + m_2}$$

  - If $m_2 \ll m_1 \Rightarrow$ the loss in energy is small
  - If $m_2 \gg m_1 \Rightarrow$ the loss is almost 100 %

- Suppose $v_{i2} = 0$; $m_1, m_2, v_{i1} \rightarrow$ known
- Collision is NOT head-on
- Unknown: $v_{f1}, v_{f2}, \theta_1, \theta_2$
- $\Rightarrow m_1 \vec{v}_{i1} = m_1 \vec{v}_{f1} + m_2 \vec{v}_{f2}$ (1)
- (1)/ox:
  $m_1 v_{i1} = m_1 v_{f1} \cos\theta_1 + m_2 v_{f2} \cos\theta_2$
- (1)/oy: $m_1 v_{f1} \sin\theta_1 = m_2 v_{f2} \sin\theta_2$
- **Elastic collision**:
  $m_1 v_{i1}^2/2 = m_1 v_{f1}^2/2 + m_2 v_{f2}^2/2 \Rightarrow$ need one more condition!



- **Totally inelastic colliding**:
  - Unknown: θ_f, v_f
  - $m_1\vec{v}_{i1} + m_2\vec{v}_{i2} = (m_1 + m_2)\vec{v}_f \quad (2)$
  - (2)/ox: $m_1 v_{i1} + m_2 v_{i2}\cos\theta_i = (m_1 + m_2)v_f \cos\theta_f$
  - (2)/oy: $m_2 v_{i2}\sin\theta_i = (m_1 + m_2)v_f \sin\theta_f$

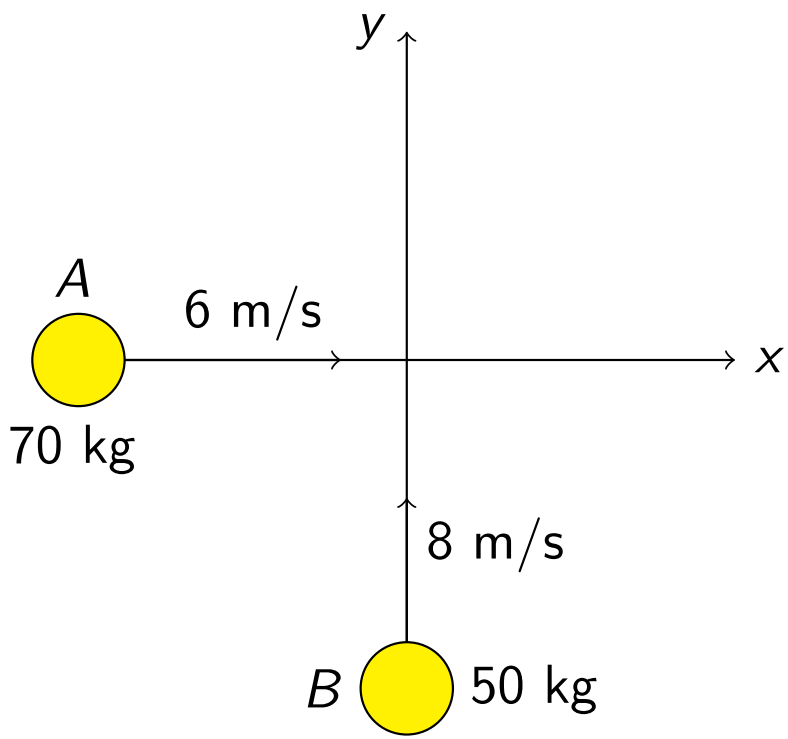


Worked example

Two people A and B of masses 70 kg and 50 kg respectively ski with velocities 6 m/s and 8 m/s along the directions as shown in the figure. They collides and hug each other after collision. What is:

(a) their velocity after collision?

(b) the fraction loss of kinetic energy?



Energy

- Energy is the capacity to do work and is measured in joules (J).
- Different forms of energy: kinetic energy, potential energy, light energy, thermal energy, electrical energy and nuclear energy.
- Kinetic energy is the energy that a body possesses due to its motion

$$K = \frac{1}{2}mv^2$$

- Potential energy is energy stored in an object due to its position and state of shape. Example: chemical, gravitational and elastic potential energy.

Energy conversion

- Energy can be converted from one form to another.
- The efficiency of a device used to convert one form of energy into another:

$$\text{Efficiency} = \frac{\text{useful energy output}}{\text{total energy input}}$$

An example of energy conservation: energy conservation during free-fall

- $v_2^2 - v_1^2 = 2\vec{a}.\vec{S};\ \vec{a} = \vec{g};\ \vec{S} = \vec{h}_2 - \vec{h}_1$
- $\Rightarrow v_2^2 - v_1^2 - -2\vec{g}.\vec{S}$

$$\Rightarrow \frac{mv_2^2}{2} - \frac{mv_1^2}{2} = -mgh_2 + mgh_1$$

$$\Rightarrow \frac{1}{2}mv_2^2 + mgh_2 = \frac{1}{2}mv_1^2 + mgh_1$$

- **Total energy** E of the mass m is conserved: $E = mv^2/2 + mgh$
  - $K = mv^2/2 \rightarrow$ **kinetic energy**
  - $U = mgh \rightarrow$ **potential energy**
  - E conserved $\Rightarrow \Delta K = -\Delta U$

Work

- Work done by a constant force

$$W = \vec{F}.\vec{s} = F.s.\cos\theta$$

- Work of the force varies with position $\vec{F} = \vec{F}(\vec{r})$

$$W = \sum_{i=1}^{N} \vec{F}_i.\Delta\vec{r}_i = lim_{N\to\infty} \sum_{i=1}^{N} \vec{F}_i.\Delta\vec{r}_i = \int_A^B \vec{F}(\vec{r}).d\vec{r}$$

- The object is acted by many forces: $W_i = \int_A^B \vec{F}_i(\vec{r}).d\vec{r}$

$$W = \sum_i W_i = \sum_i \int_A^B \vec{f}_i(\vec{r}).d\vec{r}$$

- **Work-Energy theorem**: ?? prove:

- The kinetic energy K is the energy of a body due to its motion. *Work-kinetic energy theorem:* **Work done by the net force F is changed into the kinetic energy of the body**

$$W = \Delta K$$

- Work performed by a 1-dimensional force: From $W = \int_{x_A}^{x_B} f(x)dx$, prove: $W = \Delta K$?

$$f = m\frac{d^2x}{dt^2} \Rightarrow W = \int_{x_A}^{x_B} m\frac{d^2x}{dt^2}.dx = \int_{t_A}^{t_B} m\frac{d^2x}{dt^2}.\frac{dx}{dt}.dt = \int_{t_A}^{t_B} \frac{d}{dt}\left[\frac{m}{2}\left(\frac{dx}{dt}\right)^2\right]$$

$$x(t_A) = x_A,\ x(t_B) = x_B,\ v_A = \left.\frac{dx}{dt}\right|_{t=t_A},\ v_B = \left.\frac{dx}{dt}\right|_{t=t_B}$$

$$\Rightarrow W = \frac{1}{2}m.v_B^2 - \frac{1}{2}mv_A^2 = \Delta K$$

- **potential energy** is the energy associated with a system of objects (consists of two or more objects that exert forces on one another).

Gravitational Potential Energy

- **Gravitational potential energy** is the potential energy of the object - Earth system.
- **gravitational potential energy** is defined as the product of the magnitude of the gravitational force mg acting on an object and the height y of the object

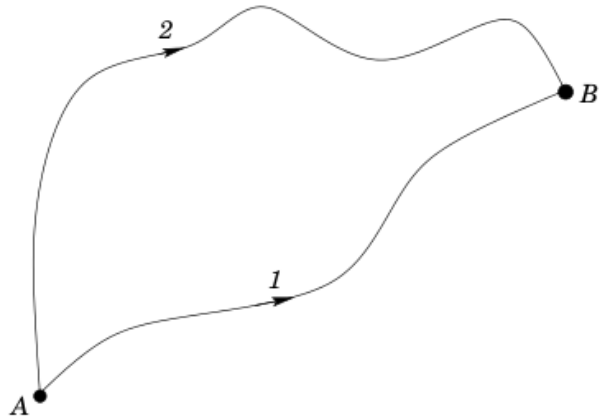
$$U_g = mgh$$

- The work done on any object by the gravitational force is equal to the negative of the change in the system's gravitational potential energy:

$$W_g = U_i - U_f = -\Delta U_g$$

- only the difference in the gravitational potential energy at the initial and final locations that matters
- Horizontal motion does not affect the value of W_g

A non-uniform force-field $\vec{f}(\vec{r})$ acts upon an object which moves along curved trajectory (from point A to point B) labeled by "path 1" and "path 2".



- Work W_1 performed by $\vec{f}(\vec{r})$ along path 1: $W_1 = \int_{A\to B}^{path1} \vec{f}.d\vec{r}$
- Work W_1 performed by $\vec{f}(\vec{r})$ along path 2: $W_2 = \int_{A\to B}^{path2} \vec{f}.d\vec{r}$
- If $W_1 = W_2 \Rightarrow \vec{f}(\vec{r})$ is a **conservative force-field**: the line-integral doesn't depend on the path taken between the end points; Ex: gravitational force.
- If $W_1 \neq W_2 \Rightarrow \vec{f}(\vec{r})$ is a **non-conservative force-field**: the line-integral depends both on the end points A, B and the path taken between them; Ex: frictional force.

Consequences of conservative and non-conservative force-field

- Work done around a close loop for:
  - Conservative force-field:

$$\Delta W = \int_{A \to B} \vec{f}.d\vec{r} + \int_{B \to A} \vec{f}.d\vec{r} = W_1 - W_2 = 0$$

  - Non-conservative force-field:

$$\Delta W = W_1 - W_2 \neq 0$$

Chapter 3: Elastic Potential Energy

- Consider a system consisting of a block plus a spring
- Force: $f = -kx$
- Work: $W = \int_{x_A}^{x_B} f(x)dx = -k\int_{x_A}^{x_B} x.dx = -[kx_B^2/2 - kx_A^2/2]$
- Potential energy: $U(x_B) - U(x_A) = -\int_{x_A}^{x_B} f(x).dx = kx_B^2/2 - kx_A^2/2$

  $\Rightarrow U(x) = kx^2/2$

  $$\Rightarrow f = -\frac{dU}{dx}$$
- In three-dimensional;

$$\vec{f} = -\left(\frac{\partial U}{\partial x}, \frac{\partial U}{\partial y}, \frac{\partial U}{\partial z}\right)$$

- Example in gravitational force-field: $U = mgz$
  $\Rightarrow \vec{f} = (0, 0, -mg)$
- Total energy of the mass: $E = K + U$

- The total mechanical energy of a system remains constant in any isolated system of objects that interact only through conservative forces.

$$K_i + U_i = K_f + U_f$$

- If there are more than one conservative force acting on an object within a system (sum over all conservative forces present):

$$K_i + \sum U_i = K_f + \sum U_f$$

Chapter 3: Work done by non-conservative force

- If there are some of the forces acting on objects within the system are **not conservative**, the total mechanical energy of the system does not remain constant. The change in mechanical energy is equal the total work done by non-conservative forces present.

Work done by an applied force

- Work-kinetic energy theorem: $W_{app} + W_g = \Delta K$
- Gravitational force is conservative $\rightarrow W_g = -\Delta U$
- $\Rightarrow \quad W_{app} = \Delta K + \Delta U$
  $\Rightarrow$ an applied force can transfer energy into or out of the system.

Work done by kinetic friction

- Kinetic friction makes total mechanical energy of the system dissipated (as heat, e.g):

$$\Delta E = \Delta K + \Delta U = \vec{F}_f.\vec{S} < 0$$

Worked example

A block of mass $m = 3\ kg$ starts at rest at a height of $h = 43\ cm$ on a plane that has an angle of inclination of $\theta = 35^\circ$ with respect to the horizontal. The block slides down the plane, and, upon reaching the bottom, then slides along a horizontal surface. The coefficient of kinetic friction of the block on both surfaces is $\mu = 0.25$. Using the *work-kinetic energy theorem*, determine how far does the block slide along the horizontal surface before coming to rest?

$$\text{Average power} = \frac{\text{Work done}}{\text{Time interval}}$$

$$\Rightarrow \text{Power} = \vec{F}.\vec{v}$$

Worked example

A 1000 kg motorcar travels at a constant speed of 30 m/s along a road with a frictional force of 100 N. What is the power generated by the motorcar? ($g = 10$ m/s^2)

A 3000 W

B 4500 W

C 300 000 W

D 450 000 W

Worked example

A car of weight 3000 N possesses an engine whose maximum power output is 160 kW. The maximum speed of this car on a level road is 35 m/s. Assuming that the resistive force (due to a combination of friction and air resistance) remains constant, what is the car's maximum speed on an incline of 1 in 20 (i.e., if θ is the angle of the incline with respect to the horizontal, then $\sin\theta = 1/20$)?